

CHG 569

Environmental Pollution Engineering

Module IX

- Environmental Ethics
 - Noise Pollution
 - Radiation Pollution
 - Indoor Pollution
- 

Environmental Ethics



The simplified definition of **Engineering**

- ❖ “the creative *application of scientific knowledge* to analyse, design, construct and operate the *products and services* of societal needs;
- ❖ with full cognizance of the environment, *sustainable development* and foremost the *safety, health and welfare* to human life.”

- ... concerning ethics.

Engineering is a tool for **living well (humans)** in the world

Engineering and the Environment

- Sources: 1880's – 1940's
 - Conservation
- Sinks: 1950's – 1980's
 - Pollution control
- Systems: 1990's – Present
 - Sustainability
 - Life cycle Analysis
 - Industrial Ecology

POLLUTION AND ENVIRONMENTAL ETHICS

- **Environmental ethics** is the part of environmental philosophy which considers extending the traditional boundaries of ethics from solely including humans to including the non-human world.

Environmental ethics is about including the rights of non-human animals in our ethical and moral values. Even if the human race is considered the primary concern of society, animals and plants are in no way less important. They have a right to get their fair share of existence.

Ethics and Morals

Environmental ethics also has to do with the study of the moral issues and decisions facing individuals and organizations.

Morals - concerned with right and wrong conduct and motives

Ethics - Philosophical study of morality

Environmental Ethics — the study of the moral issues/responsibilities concerning the environment

Ethical problems become particularly prominent when they involve a conflict of goals – see next few cases

Case 1- Built-in obsolescence

A number of decades ago a company developed and manufactured a nickel-iron battery with a 50 years lifetime. After a few years of production the patent was sold to another electric company as the batteries were “too good” and last too long.

This is one of a large number of examples of conflicts of interest between

- **the company's interest in maximizing profit by producing goods which wear out and need to be replaced regularly and**
- **the public and environmental interests of producing goods which are long lasting, thereby minimizing unnecessary resource consumption and waste generation**

In the past, things were made to last, now things are made to be replaced

Case 2- Too close for comfort

As the only engineer in the shop who has had any training in noise pollution, you have been asked, on your third day on the job to review a manufacturer's bid for noise control devices on aeration blowers for a wastewater treatment plant. After reading the bid proposal and making a few calculations, you conclude that the noise silencers will protect the workers.

But you know that the noise level for nearby neighbours will be excessive at night. (The city has no noise pollution laws.) You know that if you are asked to go back out on bid for better noise control equipment, construction of the plant will be delayed 90 days. During these 90 days untreated sewage will enter the river.

What will you recommend to your new boss?

Case 3- To Add or Not to Add

Your friend just discovered that his firm is adding nitrites and nitrates to bacon to help preserve it (*the addition of nitrite to bacon is approved by the Food and Drug Administration*). He also knows that these compounds are precursors to cancer forming chemicals that are produced in the body. On the other hand, he realizes that certain disease organisms such as those that manufacture botulism toxin have been known to grow in bacon that had not been treated.

He asks you whether he should

- (a) protest to his superiors knowing he might get fired;
- (b) leak the news to the press;
- (c) remain silent because the risk of dying from cancer is less than the absolute . . . certainty of dying from botulism.

Case 4- To dump or Not to dump

You just got a high-paying job as a process operator in a manufacturing industry. You are told to organise the clearing of an obsolete store and there you find a drum containing machine coolant, which you know to be toxic. This discussion takes place between you and your supervisor:

Supervisor: “dump half of the coolant down the drain, then dilute the rest with tap water and pour it out”.

You: “The coolant is toxic. It might cause health problems to people who live around here, even birth defects and cancers”

Supervisor: “The toxic stuff sinks to the bottom of the drum. We’ve been doing this for years and nothing’s happened”.

You: “you mean no one said anything about it”.

Supervisor: “I hope you aren’t one of those environmentalist. It’s time to get real and get on with the job”.

What do you do?

Obey your supervisor or quit, or even report to the EPA

Many environmentally related decisions
are related more to ethics than to
engineering

Decisions made in the best interest of the
environment are based on
environmental ethics.

For the 4 cases above, we might just have to look for another alternative –
a more creative solution to resolve the conflict.

Noise Pollution

NOISE POLLUTION

noise is “unwanted or disturbing sound”



- **Sources:** transportation, especially noise caused by the motor vehicles (dominant source of noise pollution); alarms; factory machinery; office equipment; construction work; power tools; barking dogs; loudspeakers; different audio systems.
- **Health effects:** stress related illnesses, high blood pressure, speech interference, hearing loss, sleep disruption, lost productivity aggression, and other countless adverse health affects.
- **Control methods:** Reduce at source or protect their ears with hearing protection (e.g., ear plugs or ear muffs)

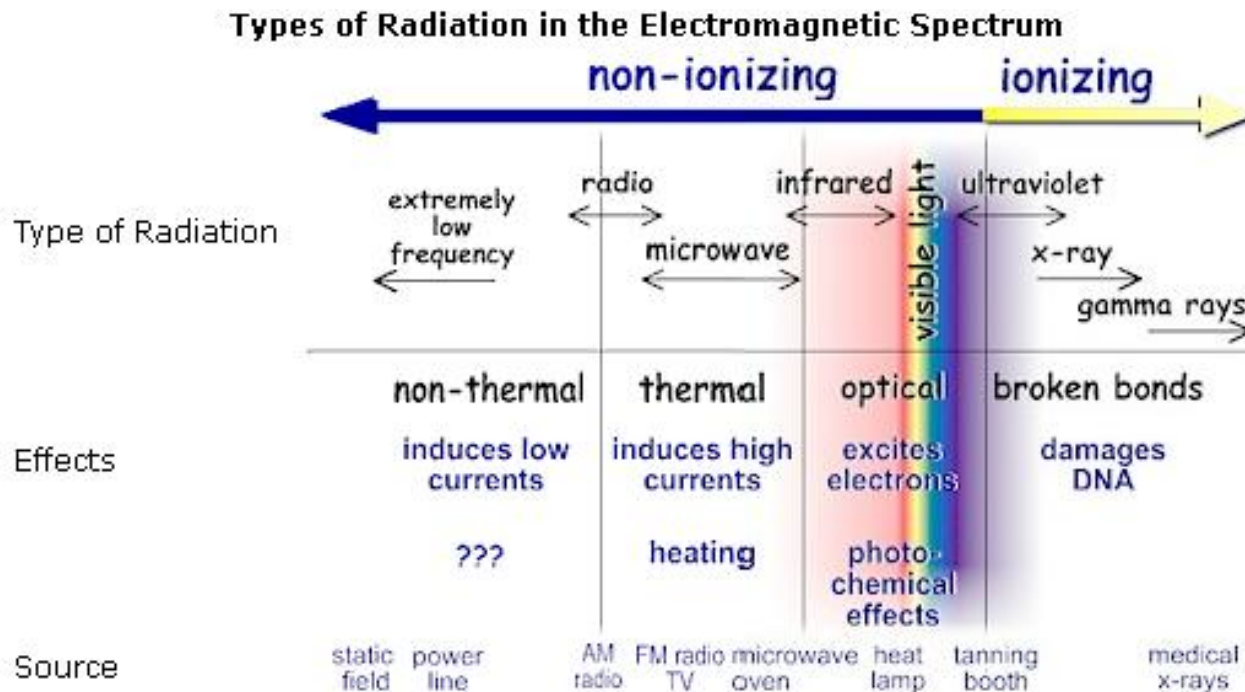
It is measured in the units of decibels and is denoted by the dB

Radiation Pollution

RADIATION (RADIOACTIVE) POLLUTION

Radiation is the emission of radiant energy that comes from a source and travels through space and may be able to penetrate various materials

2 types of Radiation - Non-Ionizing (e.g. Light, radio, and microwaves) - Ionizing



RADIATION (RADIOACTIVE) POLLUTION CONTD

Sources: Radon

— a product of the radioactive decay of Radium (naturally occurring in soil)

Nuclear power plants — Uranium mining & processing, Nuclear accidents, disposal of nuclear waste

Effects: Skin cancer, Bone cancer, swelling of bone joints, Leukemia (Blood cancer), lung cancer, genetic mutation, life span shortening, ulcer, eye problems etc

Control Methods: Avoid constructing nuclear power plants
Avoid the use of nuclear weapons
Reduce mining of uranium to a minimum
Research into proper treatment of nuclear waste
Ensure housing foundation has no cracks/leaks and proper ventilation

INDOOR AIR POLLUTION: RADON

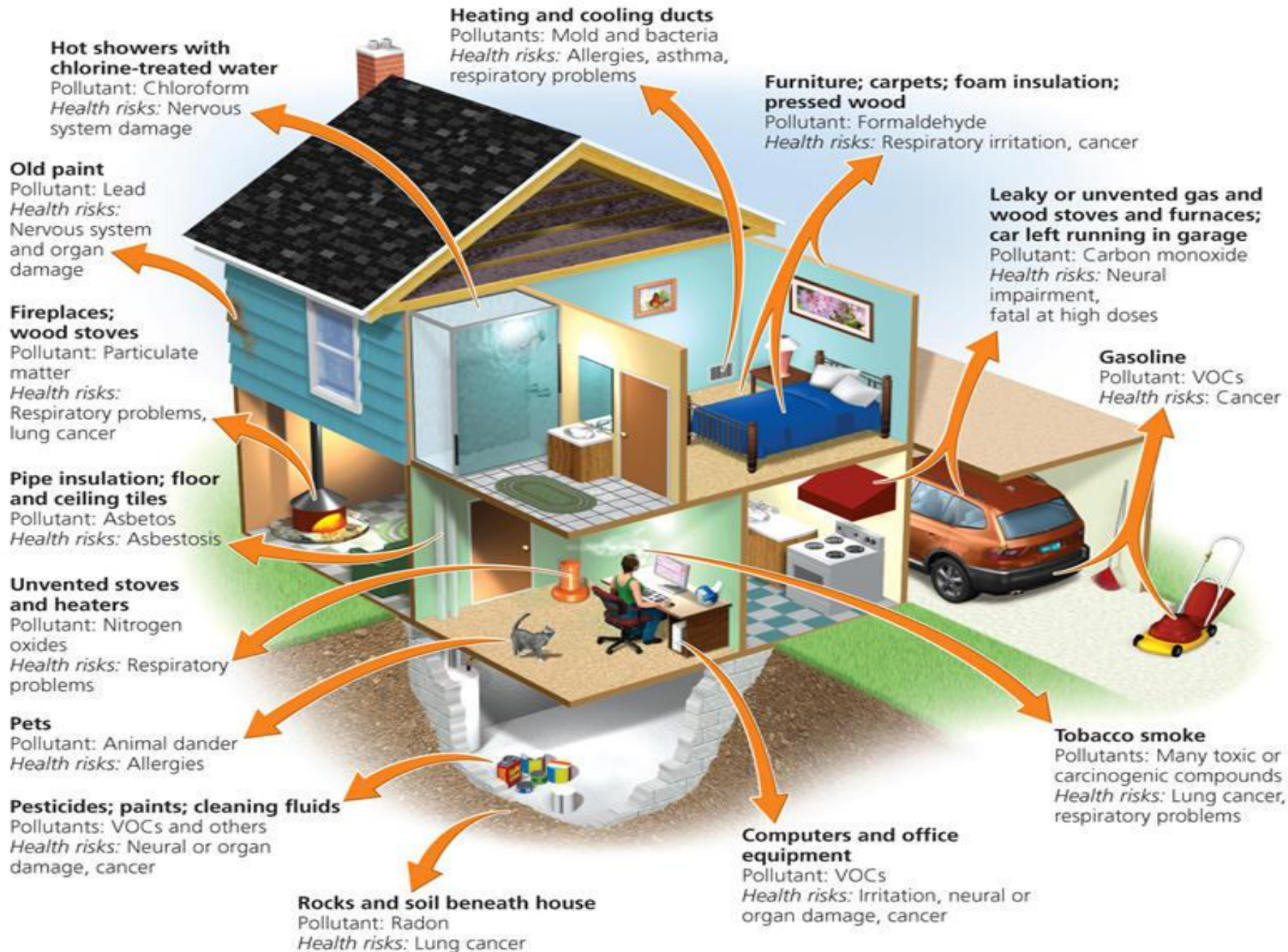
- In 1990 EPA placed indoor air pollution at the top of the list of 18 sources of cancer risk
- Indoor pollution is rated by risk analysis scientists as high-risk health problem for humans
- Radon is one of the three most dangerous indoor air pollutants, along with cigarette smoke and formaldehyde
- Radon is the second leading cause of lung cancer, after smoking
- Nearly 1 in 15 homes in the U.S. has high level of indoor radon
- The U.S. Surgeon General and EPA recommend all homes be tested for radon
- Homes with high radon level can be fixed

Radon gets in thru:

- Cracks in solid floors
- Construction joints
- Cracks in walls
- Gaps in suspended floors
- Gaps around service pipes
- Cavities inside walls
- The water supply



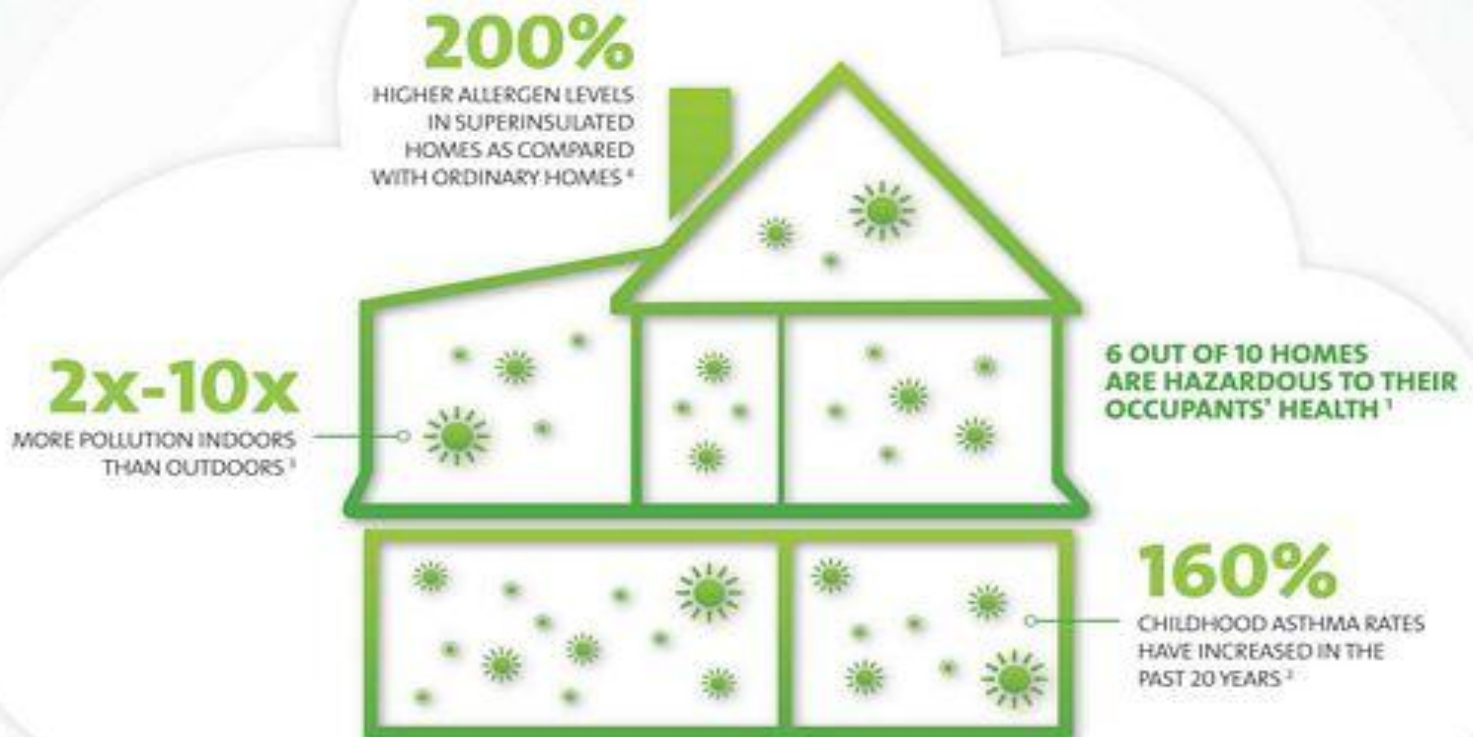
INDOOR POLLUTION



INDOOR AIR QUALITY

NEW CHALLENGES

PEOPLE ARE USING, ENCOUNTERING, INGESTING AND EXPERIENCING MORE CHEMICALS IN THEIR WORK AND HOME LIVES. IT IS CRITICAL TO UNDERSTAND THE POTENTIAL HAZARDS AND MINIMIZE THEIR IMPACT.



The Charge

*Solving our
environmental problems
will help build
a more secure and sustainable future.*

*This has become your charge and
responsibility, as a student who just
finished CHG 519 -
Environmental Pollution & Engineering
and as a future leader,*

